

Virtual Labs

For the virtual lab effort, we will use a free electronics simulation tool called “LTSpice”. We will use this tool as it is free, and can work offline, which may be of benefit for those with a less than a perfect internet connection.

“Spice” denotes the method of simulation used to explore different types of electronic circuits.

Step 1: Download the tool

Download LTSpice from (It's free!):

<https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator>

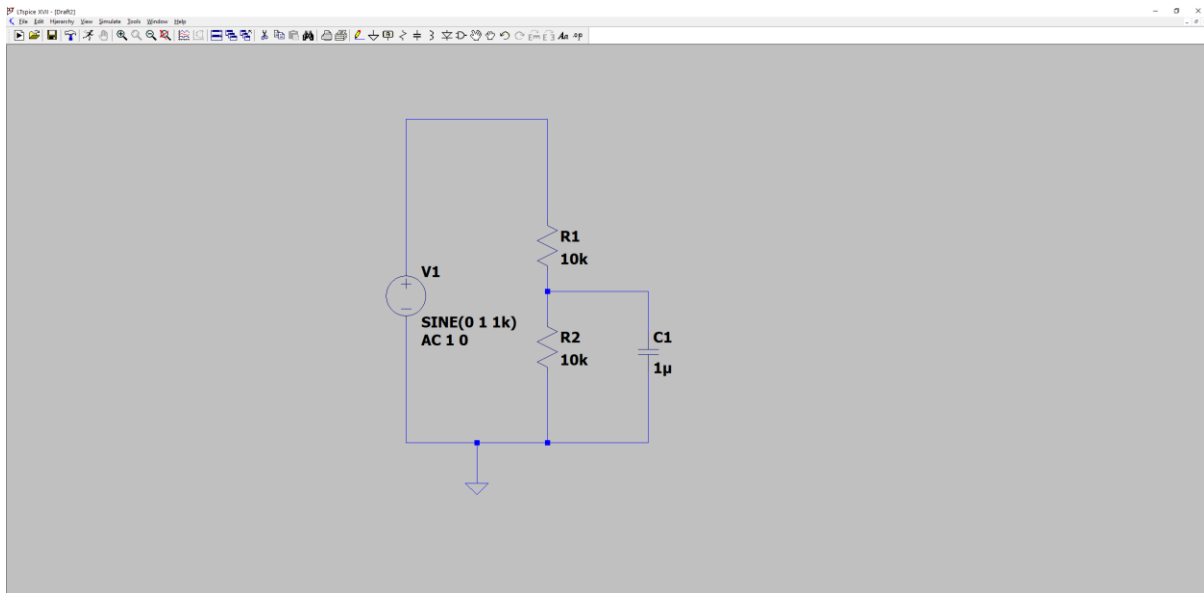
It should only take 100Mb on your computer.

Step 2: Familiarise yourself with the tool:

When you start the program, you should see a screen like the one below (except blank). Explore what each of the icons does and explore a few key controls:

- Esc: To stop your current function
- Right-click: To change the parameters of a device
- Delete: To switch to delete mode – then left-click to remove.
- Mouse wheel: Zoom in and out.

Step 3: Make an exemplar test circuit:



The circuit shown in this diagram is a potential divider. i.e. the output voltage V_{OUT} can be given by:

$$V_{OUT} = \frac{(R2 \parallel Z_{C1})}{R1 + (R2 \parallel Z_{C1})} V_1 = \frac{(10k \parallel (1/2\pi f(1\mu)))}{10k + (10k \parallel (1/2\pi f(1\mu)))} V_1$$

Where:

$$R2 \parallel C1 = \frac{Z_{C1} \cdot R2}{R2 + Z_{C1}}$$

If the frequency f is low then: $R2 \parallel C1 = R2$

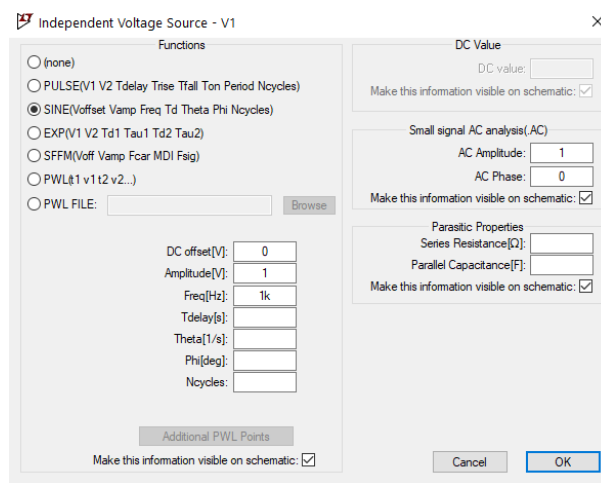
If the frequency f is high then: $R2 \parallel Z_{C1} = Z_{C1}$

Step 4: Perform some simulations

For our purposes in the virtual labs, we will perform three types of simulations:

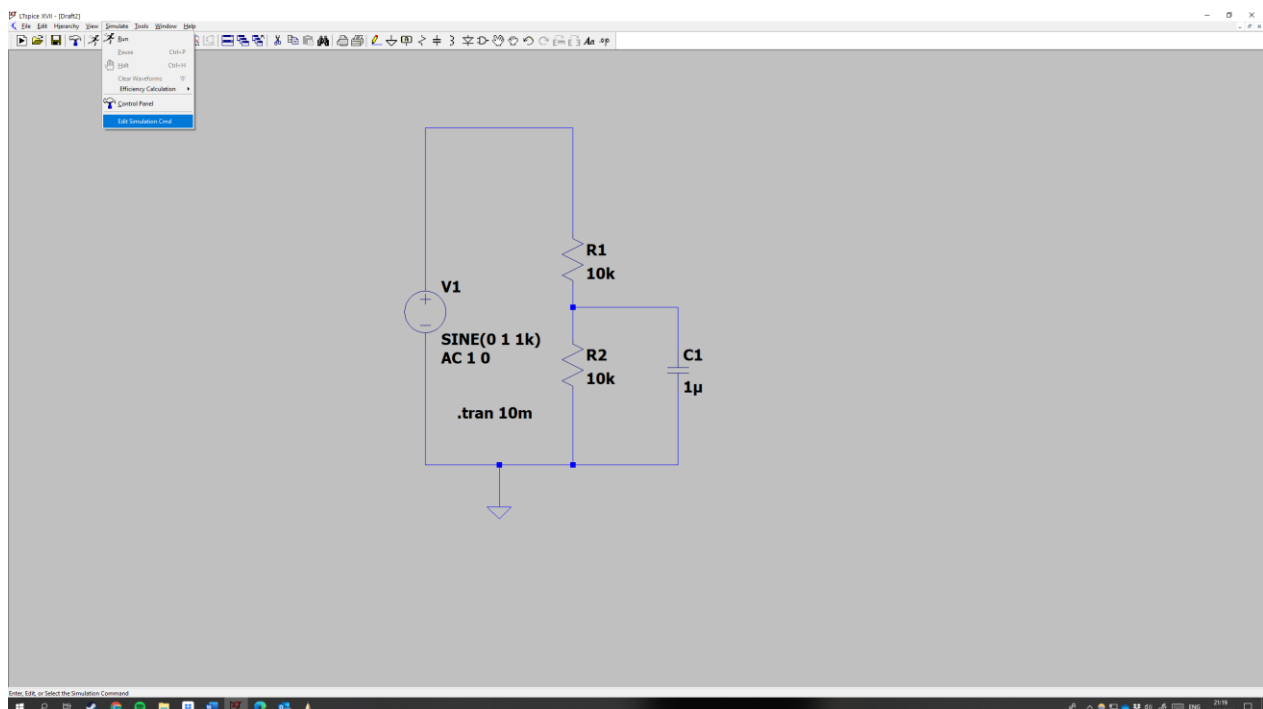
- Transient:** Analysing a point in the circuit (voltage or current) as time passes
- DC:** Analysing how a point in the circuit causes a DC change
- AC:** Performing a frequency analysis on the circuit.

As we are going to perform a transient analysis, I suggest you use the following settings for the voltage source:



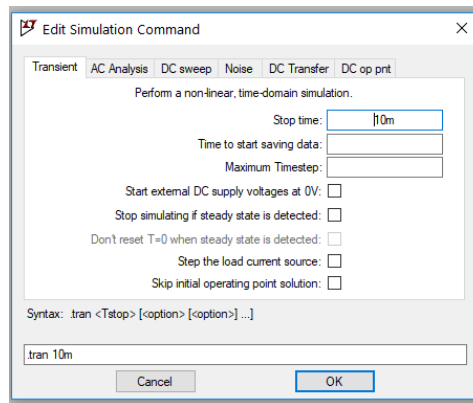
We can set the voltage source to provide a sine wave. The settings on the left: DC offset, Amplitude, Freq are for the DC and transient simulations. The small signal AC analysis is for the AC simulations.

Go to the simulation tool by clicking *simulation* -> “Edit Simulation Cmd”, then choose simulation type.

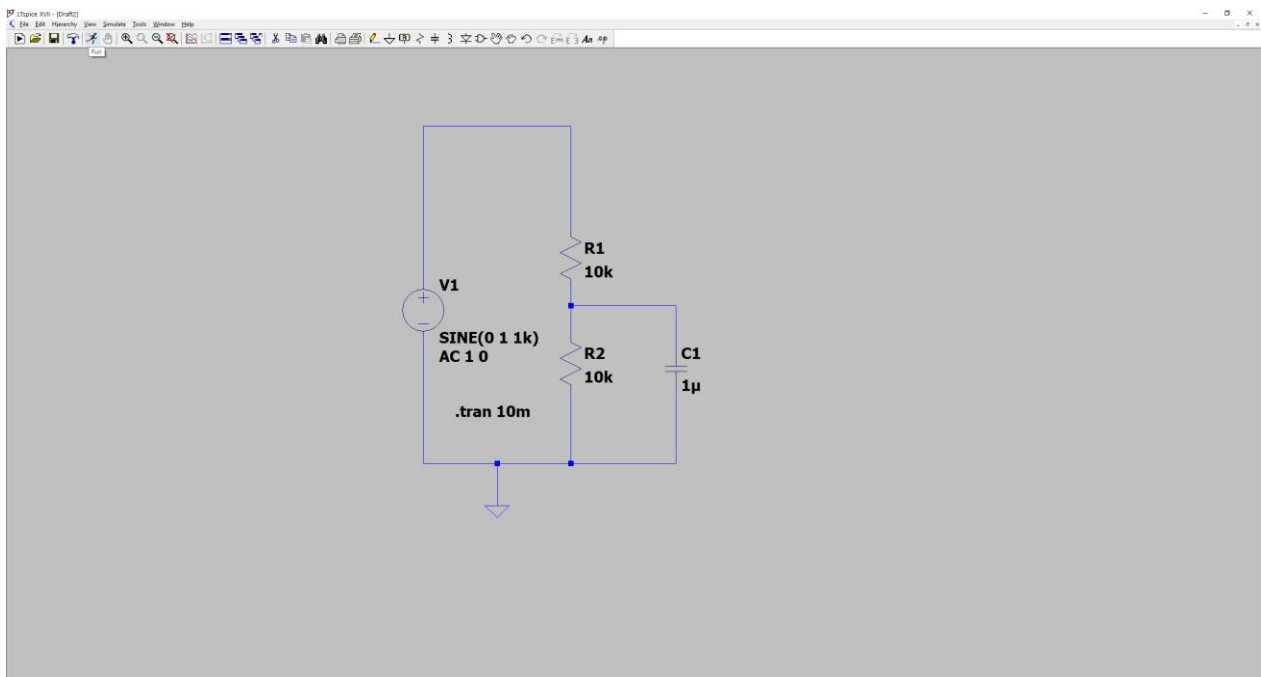


(i) Transient simulation:

Click on the transient simulation choose 10 milliseconds (10m) as the stop time, then click OK.

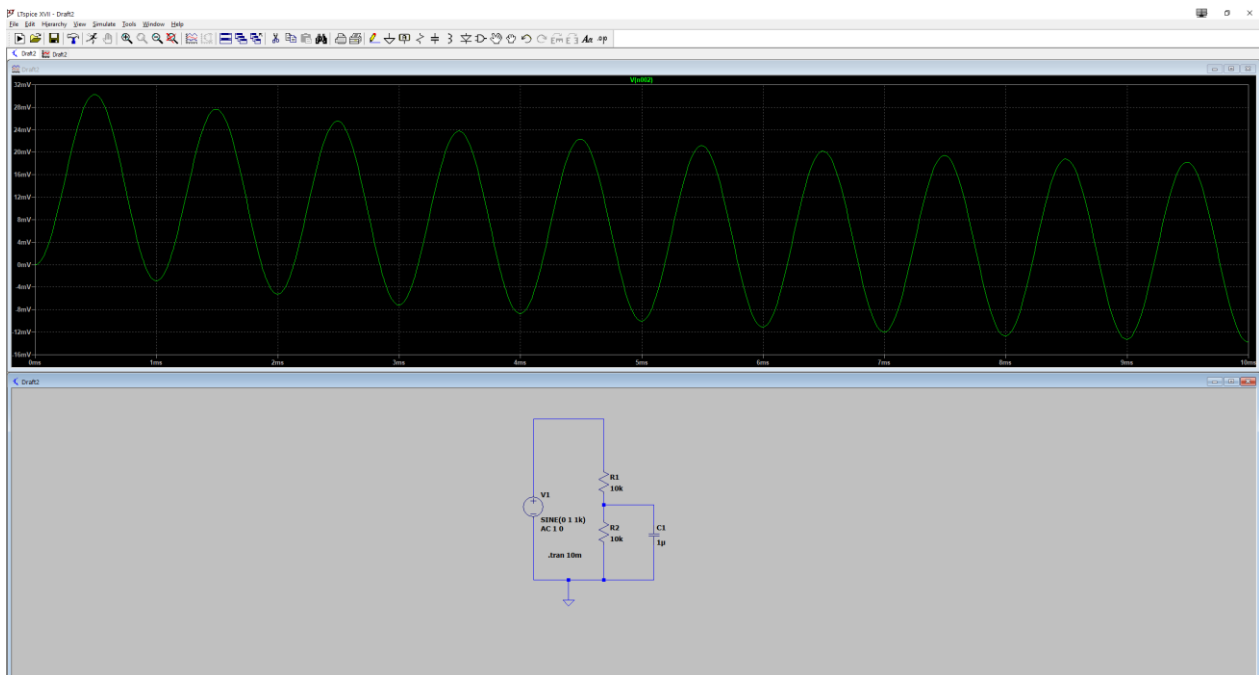


Then it the run command, or “Simulate” -> “Run” from the menu.



In order to see the results of the simulation, you can click either:

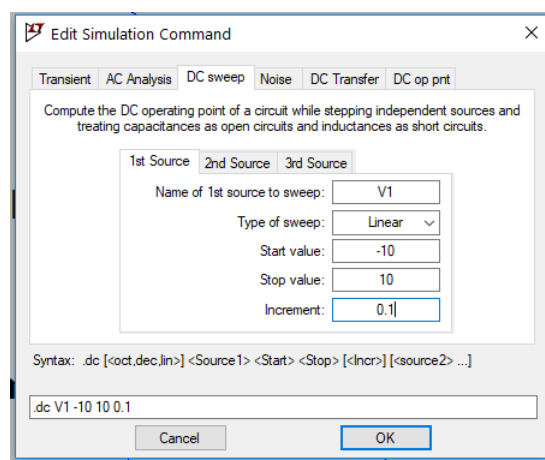
- Individual lines: (Called Nets) – which provides voltage on that line (net) compared to ground.
- Individual devices: Which provides results of the current through that device



Now try the simulation again and see what happens if you change C1 from 1μF to 1nF.

(ii) DC simulation:

Now let's try the DC simulation. Use the settings below. Note we are choosing the voltage source V1 as the element to scale (you could also choose, e.g. the resistors).



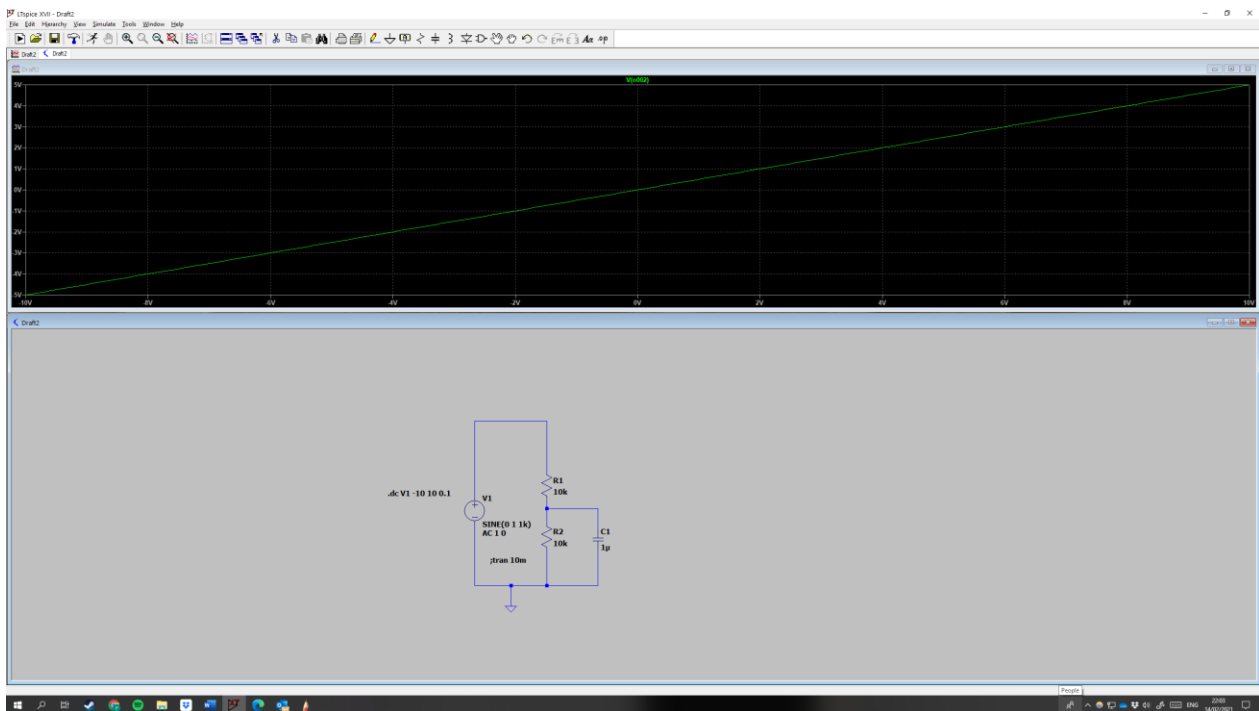
Then click run (It is best if you remove the previous simulated results first).

The result can be seen below. The x-axis provides the voltage from -10V to +10V. However the y-axis is only between -5V to 5V. This is because for a DC operation, the capacitor can be considered an infinite impedance, i.e. $R2 \parallel C1 = R2$

As such,

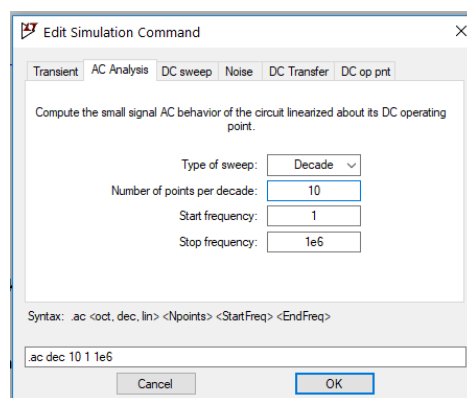
$$V_{OUT} = \frac{R2}{R1 + R2} V_1 = \frac{10k}{10k + 10k} V_1 = \frac{1}{2} V_1$$

Which is what can be seen below.



(i) AC simulation:

Now let's try the AC simulation. Use the settings below. Note, it will automatically assume the source is the one providing “AC settings”.



Then click run (It is best if you remove the previously simulated results first)

In this case, we are plotting something called a “Bode plot”. The x-axis is frequency over a logarithmic scaling. The y axis is in dB = 20 log (V).

Note that the value goes down after a certain frequency (~30Hz) This is called the 3dB cutoff where the amplitude drops by 50% (3dB).

